

Responses to the Reviewers' Comments on the manuscript with ID TCOM-TPS-25-2414

Model-Based Beam-Steered Optical Wireless Positioning with Single-LED Single-Photodiode for 3D Localization

by Kevin Acuna-Condori, Bastien Béchadergue, Hongyu Guan, and Luc Chassagne

We would like to thank the reviewers and the editor for their constructive comments. They have been very helpful in the revision of this paper and allowed us to improve the technical content and presentation quality. We thus hope that we have taken into consideration and complied with all the comments and suggestions.

In this document, we provide detailed explanations of the changes made in the revised version of our article to address these comments. Each comment is reminded in **red** and is followed by our response, in black font. The changes made to the revised manuscript are mentioned in **blue**. Please also note that, unless stated otherwise, all page, section, figure, table, and reference numbers, as well as acronyms refer to the revised manuscript.

Responses to the Editor

The paper studies a timely and interesting problem and presents a potentially valuable model-based approach for 3D localization using a single beam-steered LED and a single photodiode. The work has merit, but in it requires major revision.

In preparing the revision, the authors must address the following points:

- 1. The manuscript must resolve the inconsistency between the claimed capability of operation without receiver rotation and under arbitrary receiver orientation and the actual system model and estimation framework, which rely on a fixed known receiver normal and a beam-aligned distance-recovery step. This is a central issue and must be corrected both technically and in the wording of the claims.*
- 2. The paper must substantially improve the clarity of the formulation and derivations, including the motivation, description of the proposed procedure, notation, estimator development, and the link between the steering parameters and the optimization criterion.*
- 3. The validation **must** be strengthened to better support the claimed practical relevance of the method. The revised version **must** include more convincing benchmarking and a clearer treatment of robustness and implementation-related issues raised by the reviewers.*

We would like to thank the editor for his overall assessment of our work and for this summary of the reviewers' most important comments, which we address in detail later in this document.

From a general perspective, **the main change we made to our work to address these comments was to give greater emphasis to the direction-finding step, which is entirely independent of the receiver's orientation as initially claimed, compared to the distance recovery and therefore positioning step itself, which requires rotation of the receiver, as the reviewers noted.** This major change led to a number of modifications in this revision, which are summarized here:

- **Complete rewriting of the introduction (Section I)** with clear logical flow, systematic literature review, and explicit positioning of direction finding as the primary contribution.
- **Direction Error Bound (DEB):** A new Cramér–Rao-based bound dedicated to the direction-finding stage (Section III) is now derived, with Schur-complement profiling of the nuisance parameter—providing the first information-theoretic benchmark specific to single-LED direction estimation on $\mathbb{S}^2$.
- **Comprehensive DEB-based optimization (Section IV):** Seven DEB-optimized orientation sets ($K = 3$ to 9) are computed, with root mean square (RMS)-DEB decreasing monotonically from 1.30° to 0.36° (Table III). DEB violin comparison across K with 50 random realizations per value (Fig. 2) is provided, along with spatial DEB heat maps (Fig. 3), RMS-DEB / position error bound (PEB) vs. K for three semi-angles at half power (Fig. 4), and RMS-DEB/PEB vs. SNR confirming the theoretical $1/\sqrt{\text{SNR}}$ scaling (Fig. 5).
- **Dedicated direction-finding evaluation (Section VII-A):** Standalone assessment using angular error over $1,792$ positions $\times$ $1,000$ Monte Carlo trials (≈ 1.8 million evaluations per estimator) is provided, yielding sub-degree RMS error for all estimators (Table IV, Fig. 6).
- **Formal proof of receiver-orientation independence (Section VI-C)** for all three estimators is provided and validated numerically with 100 random tilts per position ($< 3\%$ degradation; Table V, Fig. 6).
- **Computational latency analysis (Section VII-A-3, Fig. 7)** is refined, and shows that generalized least squares (GLS) / weighted least squares (WLS) achieve μs -level inference (0.025/0.018 ms), whereas nonlinear least square (NLS) achieves sub-millisecond latency (0.65 ms) via Levenberg–Marquardt.
- **Estimator performance vs. SNR (Section VII-C, Fig. 10):** Full Monte Carlo characterization of all estimators across $\text{SNR} \in [10, 50]$ dB (1,792 positions, 1,000 trials per point, ≈ 16.1 million evaluations per method) is provided, confirming that all estimators follow the $1/\sqrt{\text{SNR}}$ slope and quantifying the accuracy–latency trade-off.
- **Updated Table I** with two-stage entry and clarifying footnote.
- **Note below Table IV** clarifying PEB spatial statistics.

Responses to Reviewer 1

The authors present a single beam-steered LED single-PD OWP localization architecture, and derived the PEB. They discuss the selection of the orientation sets. They develop direction finding methods and demonstrate their feasibility by numerical simulation.

We would like to thank the reviewer for the positive assessment of our work and for the constructive comments that have helped us improve the clarity of the paper. We are pleased that the reviewer found the architecture, the PEB derivation, and the direction-finding methods to be sound and demonstrated through simulation. We hope that the responses below satisfactorily address all of the remaining comments from Reviewer 1.

Some minor issues need to be addressed before acceptance.

1. L37-40, left, page 1: please correct the sentence “there is..., which ..., such as ..., are widely studied ...”

The Introduction has been completely rewritten in the revised manuscript. The problematic sentence has been replaced by a grammatically correct and clearer formulation in the first paragraph of Section I:

“Dedicated indoor positioning systems (IPS) are therefore required, and radio-frequency (RF) techniques leveraging existing communications infrastructure—such as Wi-Fi, Bluetooth Low Energy (BLE), ultra-wideband (UWB), radio-frequency identification (RFID), and emerging 5G positioning—are widely studied and deployed.”

2. L51, left, page 2: “and” should be added before “silicon-photonics”.

The sentence has been rewritten for clarity (see Section I, sixth paragraph).

3. Table II: why do you use the z range of $[0, 1.2]$ instead of a larger one such as $[0, 2]$? Which half-power semi-angle of the 3 values is selected in “optimization”?

Regarding the **z -range**: The transmitter is ceiling-mounted at $z = 2$ m, so $z = 0$ corresponds to the floor and $z = 1.2$ m corresponds to a receiver held at chest height. Extending the range to $z = 2$ m would place the receiver at the ceiling (transmitter height), which is physically unrealistic for a handheld or body-mounted device. Moreover, at very short transmitter–receiver separations (z close to 2 m), the incidence angle ψ approaches the field-of-view (FOV) limit, the Lambertian model becomes ill-conditioned, and the Fisher information matrix (FIM) degenerates. The range $[0, 1.2]$ m therefore covers the practical receiver-height envelope for indoor pedestrian scenarios.

Regarding the **semi-angle at half-power**: The genetic algorithm (GA) optimization was run independently for each of the three values $\Phi_{1/2} \in \{30^\circ, 45^\circ, 60^\circ\}$ listed in Table II, producing three sets of DEB-optimized orientations. Table III then lists the resulting orientations when $\Phi_{1/2} = 45^\circ$, whereas Fig. 4 (RMS-DEB/PEB vs. K) shows the results for all three. Based on this comparison, $\Phi_{1/2} = 45^\circ$ was selected for the subsequent simulations because it offers a favorable trade-off between DEB performance and widespread availability in commercial LEDs.

The text accompanying Fig. 4 in Section IV now states: “ $\Phi_{1/2} = 45^\circ$ achieves competitive performance starting at $K = 3$ and, given its widespread use in commercial LEDs, is adopted for the subsequent simulations”. The titles of Table III, Fig. 2 and Fig. 3 have also been modified to include the value of $\Phi_{1/2}$ it corresponds to (i.e. $\Phi_{1/2} = 45^\circ$), as well as the sentence introducing Table III in the main text (fourth paragraph of Section IV).

1. *Fig 2: how many random orientation sets are used?*

50 random orientation sets are generated per K value. For each random set, the DEB is evaluated over all 1,792 testbed positions, and the resulting spatial distribution is aggregated into the orange violin/box plot. This number provides a statistically meaningful characterization of the random baseline while keeping computational cost manageable.

The caption of Fig. 2 and the accompanying text in Section IV have been updated to explicitly state the number of random realizations. The caption now reads: “Green: GA-optimized orientation set; orange: randomly selected set”—the text below the figure clarifies that 50 random sets are used per K .

2. *Fig 3b: please clarify what you mean by “random configuration”. Is it one of the sets other than the optimized one? How do you select it?*

The “random configuration” in Fig. 3(b) is **one single randomly generated orientation set** for $K = 5$, where each tilt–azimuth pair (θ_i, φ_i) is drawn uniformly from the same search range as the GA $(\theta_i \in [0^\circ, 80^\circ], \varphi_i \in [0^\circ, 360^\circ])$. It was selected to represent a typical (non-optimized) deployment for visual comparison with the GA-optimized set in Fig. 3(a). The purpose is to illustrate the spatial DEB pattern, not to provide a statistical comparison (which is done by Fig. 2 with 50 random sets).

The caption of Fig. 3 has been updated to clarify: “(b) randomly selected set” now explicitly indicates that this is a single set with uniformly drawn tilt–azimuth pairs.

3. *Fig 4: I didn’t get why 60 deg half-power semi-angle has larger PEB. Doesn’t higher signal strength reduce PEB?*

This is an insightful question. The intuition that “more power $\rightarrow$ lower PEB” holds for a *fixed* beam direction, but our system relies on *angular discrimination* between K orientations. The key is the Lambertian order m :

- $\Phi_{1/2} = 45^\circ \rightarrow m \approx 2.0$ (more directive beam),
- $\Phi_{1/2} = 60^\circ \rightarrow m \approx 1.0$ (wider, less directive beam).

The DEB and PEB depend on the **gradient of the received power with respect to the direction**, which appears in the FIM as the term $m \cdot Q_i^{m-1}$ in (16). A higher Lambertian order m (more directive beam) produces steeper power variations as the beam is steered, amplifying the contrast between orientations and thus increasing the Fisher information. A lower m (wider beam) flattens the power profile, reducing the angular sensitivity: the K received powers become more similar regardless of the receiver direction, and the FIM shrinks.

In other words, while a wider beam distributes *more total power* across the room, it provides *less directional information* per orientation. Since the DEB/PEB depends on the *sensitivity* of power to

direction (not on the absolute power level), a narrower beam ($\Phi_{1/2} = 45^\circ$) outperforms a wider one ($\Phi_{1/2} = 60^\circ$) for direction finding and positioning.

The text accompanying Fig. 4 in Section IV now explains: “Wider beams ($\Phi_{1/2} = 60^\circ\text{--}75^\circ$) incur higher bounds due to the reduced Lambertian directivity”.

4. L49, left, page 10: correct the sentence “... apply the solves needed...”

The section containing this sentence has been rewritten; the issue no longer exists in the revised manuscript.

Responses to Reviewer 2

This paper presents a model-based 3D indoor positioning architecture that employs a single beam-steered LED and a single PD. Spatial hardware redundancy is replaced by temporal beam steering over K distinct orientations. The authors derive the PEB and attempt to optimize the orientation set using GA. Three estimators are then developed, namely the NL, an efficient GLS, and a lightweight WLS. Simulation results suggest that the proposed estimators can achieve reasonable performance. However, the paper contains several significant issues that prevent its acceptance in its current form.

We sincerely thank the reviewer for the thorough evaluation of the whole paper, and for all the constructive comments below, which have helped us improving both the clarity and the quality of our work. We hope that the detailed responses and the corresponding revisions adequately address all of Reviewer 2's concerns.

- 1. The motivation of the paper is not sufficiently clear. Although many related works are cited in the Introduction, the presentation lacks a clear logical flow, and the relationships among the cited works are not well established. The Introduction should be reorganized to clearly highlight the problem being addressed, the limitations of existing approaches, and the necessity of the proposed solution.*

We have **completely rewritten** Section I following a structured progression: (1) GNSS limitations indoors and the need for indoor positioning systems (IPS); (2) optical wireless positioning (OWP) advantages (spatial confinement, immunity to radio frequency interference, deterministic line of sight (LOS) channels); (3) model-based vs. data-driven paradigms; (4) multi-LED paradigm limitations (need for 3–4 visible LEDs, deployment cost); (5) single-LED state of the art via Table I showing all compensate at the receiver side; (6) beam steering as an alternative strategy shifting diversity to the transmitter; (7) core concept paragraph explaining our approach before listing contributions; (8) five numbered contributions and paper organization.

The revised Introduction now explicitly positions **direction finding as the primary contribution**, with five clearly delineated items: (i) the single-LED/single-photodiode (PD) framework with $\mathbf{n}_r$ -independence; (ii) DEB and PEB analysis with identifiability conditions; (iii) GA-based orientation-set design; (iv) closed-form GLS/WLS estimators; and (v) the NLS estimator. These contributions are supported by nine new figures (Figs. 2–10), three new tables (Tables III–IV), and Monte Carlo campaigns totaling nearly 1.8 million evaluations per estimator.

The entire Introduction (Section I) has been rewritten in the revised manuscript with the logical flow and contribution structure described above.

- 2. The core concept and steps of the proposed method are not adequately explained before listing the contributions.*

We agree. In the revised manuscript, two paragraphs immediately before the contributions list explain: (i) the core idea of steering a single LED through K orientations while a PD at an arbitrary fixed orientation collects K RSS measurements; (ii) the ratio-based formulation that cancels unknown distance, transmitted power, and all receiver-orientation-dependent terms; (iii) that direction finding is mathematically independent of $\mathbf{n}_r$; (iv) the extension to 3D via cooperative

distance recovery. A second paragraph clarifies that direction finding is the primary contribution enabling beam alignment for subsequent communication or ranging.

Two new paragraphs added in Section I before the contributions list (beginning “Building on these observations...”).

3. *The description of the "Distance Recovery" stage in Section II-B2 is confusing, particularly with respect to the so-called "beam-forming symmetry." The discussion appears to imply that the receiver must also be oriented as $\mathbf{n}_r = -\hat{\mathbf{n}}_d$, which contradicts the claim in the Abstract that the system uses a single PD without receiver rotation.*

We would like to thank the reviewer for this comment, which highlights an ambiguity that was indeed present in the original version of the article and which, we hope, has now been resolved. More precisely, the original manuscript did not adequately distinguish the two stages:

- **Stage 1 (Direction Finding):** The PD remains at an arbitrary fixed orientation throughout K measurements. All estimators are therefore $\mathbf{n}_r$ -independent (as proven in Section VI-C), no receiver movement is required. The claim that no receiver rotation is needed was thus referring to this first stage.
- **Stage 2 (Distance Recovery):** After direction finding, one single cooperative reorientation ($\mathbf{n}_r = -\hat{\mathbf{n}}_d$) is indeed required to perform ranging while maximizing SNR. It may be noted that this is a single discrete reorientation, and not a continuous rotation as in [20], [24], [27]. The reviewer is therefore right to note that this second stage contradicts the initial “no receiver rotation” claim.

To avoid any ambiguity, the revised Abstract now explicitly describes the two-stage process: the system “first estimate[s] the LED-to-PD direction, and then recover[s] the distance from a single cooperative beam-aligned measurement”. This transparent wording makes clear that the direction-finding stage requires zero receiver movement, while the distance-recovery step involves one cooperative reorientation. The Introduction (Section I) further states “the direction-finding stage requires zero receiver movement, and the entire beam-steering complexity resides in the infrastructure”. The former reference to “beam-forming symmetry” has been replaced by a transparent description of cooperative alignment.

Section II-B2 has therefore been rewritten, and a new paragraph after (9) explicitly describes the single cooperative reorientation. Table I has also been updated with two-row entry (Arbitrary/Controlled) and an explanatory footnote.

4. *Several symbols and derivations lack clarity:*

- a) *In $\mu_i(r)$ under (10), the dependence of $h_{LOS,i}(r)$ on r is unclear, since Eqs. (2) and (3) do not explicitly involve r .*

While (2)-(3) express the channel gain via $\cos \phi_i$, $\cos \psi$, and d , all depend on $\mathbf{r}$ through $\mathbf{d} = \mathbf{r} - \mathbf{t}$. We added Section III-A (“Direction-Finding Observation Model”) showing explicitly that $\mu_i = \eta \cdot Q_i^m$ (11), where $Q_i = \mathbf{n}_{t,i} \cdot \mathbf{n}_d$ depends on $\mathbf{r}$ through $\mathbf{n}_d = (\mathbf{r} - \mathbf{t})/||\mathbf{r} - \mathbf{t}||$, and $\eta = C \cdot \cos \psi / d^2$ depends on $\mathbf{r}$ through d and $\cos \psi$.

A new Section III-A has therefore been added with (10)-(13) tracing all dependencies on $\mathbf{r}$.

b) Under the Gaussian noise assumption, the FIM can be directly written in the form of (16) following standard results (e.g., Kay). Therefore, Sections III-A and III-B appear unnecessary.

We agree and have streamlined the derivation. We now directly apply the Slepian-Bangs formula (citing Kay [42]) to write the FIM in (15). The intermediate log-likelihood/score steps have been removed. The current Sections III-A and III-B are concise: Section III-A introduces the observation model and nuisance parameter η (needed for the Schur complement), whereas Section III-B defines the specific gradient vectors for our Lambertian parameterization.

Equation (15) now directly cites [42], and intermediate derivation steps have been removed.

c) The vectors $\mathbf{u}_x$, $\mathbf{u}_y$, and $\mathbf{u}_z$ appearing above (26) are not defined.

This has been corrected. In Section IV, immediately before (23), the definition now reads: “where $\mathbf{u}_x = [1,0,0]^T$, $\mathbf{u}_y = [0,1,0]^T$, $\mathbf{u}_z = [0,0,1]^T$ are the standard Cartesian basis vectors”.

d) The derivations of the GLS and WLS estimators are not sufficiently explained, making it difficult to follow the reasoning.

We added explanatory paragraphs: (i) after (30) explaining that the power ratio cancels d and $\cos \psi$, yielding $\beta_i = \cos \phi_i / \cos \phi_1$ depending only on LED orientations and displacement direction; (ii) after (32) providing geometric interpretation (each constraint defines a hyperplane through the origin; $K - 1$ hyperplanes intersect at the true direction); (iii) in Section VI-B explaining when WLS approximates GLS well vs. when full GLS is needed.

Explanatory paragraphs have thus been added after (30), (32), and in Section VI-B.

5. There is a major inconsistency between the claims in Table I and the system model. Table I states that the receiver orientation is "arbitrary," whereas Section II-A explicitly assumes a fixed receiver orientation, i.e., a single PD with normal vector $\mathbf{n}_r = [0,0,1]^T$. Moreover, the derivations in Eq. (22b) and the subsequent linear estimators rely on a known $\mathbf{n}_r$. If the receiver orientation is truly arbitrary (i.e., unknown), the problem becomes a 6-DoF estimation problem, which is not supported by the current formulation or derivations.

We thank the reviewer for this critical observation. The apparent inconsistency arises from two aspects insufficiently distinguished in the original:

1. **$\mathbf{n}_r = [0,0,1]^T$ is a simulation choice, not a model assumption.** Section II-A now states: “For concreteness, we set $\mathbf{n}_r = [0,0,1]^T$ in the simulations. As shown in Section VI-C, all proposed direction estimators are independent of $\mathbf{n}_r$; therefore, the direction-finding results hold for any receiver orientation”.
2. **The direction-finding stage is provably $\mathbf{n}_r$ -independent.** A new Section VI-C (“Receiver-Orientation Independence”) formally proves this:
 - For GLS/WLS: the power ratios $\beta_i = \left(\frac{\mu_i}{\mu_1}\right)^{1/m} = Q_i/Q_1$ cancel the common factor $\eta = C \cdot \cos \psi / d^2$ exactly. Since $\mathbf{a}_i$, Σ_β , $\mathbf{M}_{\text{GLS}}$, and $\mathbf{M}_{\text{WLS}}$ depend only on $\{\beta_i\}$ and $\{\mathbf{n}_{t,i}\}$, the estimates are $\mathbf{n}_r$ -independent.

- For NLS: the normalized targets $p_i = Q_i^m / Q_{\max}^m$ cancel η through the free scale parameter.
- 3. **The problem is not 6-DoF.** Because ratio-based formulations eliminate all receiver-dependent factors, direction finding estimates only 2 DoF (direction on $\mathbb{S}^2$). The receiver orientation need not be estimated.
- 4. **Numerical validation (Section VII-A-2):** Monte Carlo under random tilts (half-normal $\sigma = 5^\circ$, $\max = 30^\circ$) confirms all estimators degrade by less than 3% (Table V, Fig. 6).

Section II-A has thus been updated, and Section VI-C has been added with formal proof. Table I has also been updated with two-row entry (Arbitrary/Controlled) and footnote. Finally, Section VII-A-2 provides Monte Carlo validation.

- 6. *The tilt-azimuth pairs do not explicitly appear in the system model, and their relationship with the CRLB is unclear. Consequently, it is not evident how these angles are optimized using GA.*

The connection is now explicit through the following chain: (i) (23) maps (θ_i, φ_i) to $\mathbf{n}_{t,i}$; (ii) the FIM (15)-(16) depends on $\{\mathbf{n}_{t,i}\}$ through $Q_i = \mathbf{n}_{t,i} \cdot \mathbf{n}_d$; (iii) the DEB (18) is obtained via the Schur complement of the FIM; (iv) the GA objective (22) minimizes the RMSE of the DEB with decision vector $x = [\theta_1, \varphi_1, \dots, \theta_K, \varphi_K]^T$. Table II summarizes all parameters including search ranges.

The effectiveness of this optimization is demonstrated quantitatively: Table III reports seven DEB-optimized orientation sets ($K = 3$ to 9), with the RMS-DEB decreasing monotonically from 1.30° ($K = 3$) to 0.52° ($K = 5$) to 0.36° ($K = 9$). Figure 2 further validates the GA by comparing each optimized set against 50 random realizations per K value, showing that the GA reduces the median DEB by 49%–72% across all K , with consistently tighter interquartile ranges. Figure 3 provides spatial heat maps confirming that the optimized set yields uniform, low-error coverage, whereas random orientations produce irregular and larger error patterns.

Section IV now includes (22) with an explicit objective function. Table III reports all optimized sets with achieved RMS-DEB (when $\Phi_{1/2} = 45^\circ$). Finally, Figs. 2–3 provide statistical and spatial validation.

- 7. *The orientation set optimization involves $2K$ unknowns (tilt-azimuth angles), while only K measurements are available. This results in an underdetermined problem, and the solution obtained by GA may not be reliable.*

We thank the reviewer for this comment but respectfully disagree, as it seems to conflate two fundamentally different problems:

1. **Offline orientation-set design (GA):** This is a system-design problem, not an estimation problem. The $2K$ variables define the orientation set; for each candidate, the DEB is evaluated analytically over 1,792 positions using the closed-form FIM in (15). No measurements are involved, only theoretical bounds. The GA optimizes a deterministic scalar objective (22) mapping $2K$ reals to a cost value, which is a standard global optimization problem.

2. **Online direction estimation:** Once the set is fixed, the direction-finding stage estimates 2 DoF (direction on $\mathbb{S}^2$) from K power measurements. For $K \geq 3$, this is overdetermined (K measurements, 2 unknowns).

The reliability of the GA solution is demonstrated empirically. Table III reports seven independently optimized orientation sets ($K = 3$ to 9, with $\Phi_{1/2} = 45^\circ$), all exhibiting physically interpretable patterns (e.g., $K = 4$ yields four orientations at 29.9° tilt uniformly spaced in azimuth, i.e., the intuitively optimal symmetric configuration). Figure 2 compares each GA-optimized set against 50 random realizations, confirming that the optimized sets consistently reduce the median DEB by 49%–72% and yield tighter distributions across all K . These results provide strong evidence that the GA converges to high-quality solutions despite the non-convex landscape.

A paragraph in Section IV now states: “We emphasize that the GA optimization is an offline system-design problem, not an online estimation problem. [...] Once fixed, the online direction-finding stage uses K power measurements to recover the 2-DoF direction on $\mathbb{S}^2$, which is overdetermined for $K \geq 3$.”

8. *It is unclear whether the position vector $\mathbf{r}$ is assumed to be fixed when optimizing the orientation set. If $\mathbf{r}$ is unknown, a joint optimization over both $\mathbf{r}$ and the tilt-azimuth pairs should be formulated.*

In the GA optimization, $\mathbf{r}$ is not fixed. The GA objective (22) evaluates the RMSE of the DEB over all 1,792 testbed positions. The position $\mathbf{r}$ sweeps the entire spatial grid, and the GA seeks an orientation set minimizing the average DEB universally. A joint optimization over ($\mathbf{r}$, orientations) would not add value because the orientations must be fixed before any measurement, as they define the illumination pattern for all future positions.

Section IV now states: “The position $\mathbf{r}$ is therefore not fixed but sweeps the spatial grid during fitness evaluation; the resulting orientation set is position-independent and can be deployed universally”.

9. *The results shown in Fig. 9 lead to essentially the same conclusion as those in Fig. 2 (Optimized), making Fig. 9 redundant.*

In the revised manuscript, the figures have been substantially restructured to cover both the theoretical analysis and the estimator evaluation. The ten figures in the revised paper serve clearly distinct purposes:

- **Fig. 1 (System geometry):** Single beam-steered LED / single-PD OWP geometry showing the transmitter at the ceiling and the PD at the receiver.
- **Fig. 2 (DEB violin):** Statistical distribution of the theoretical DEB across $K \in \{3, \dots, 9\}$, GA-optimized vs. 50 random realizations per K , when $\Phi_{1/2} = 45^\circ$.
- **Fig. 3 (DEB heat map):** Spatial DEB distribution at $z = 0.8$ m (optimized vs. random, $K = 5$), when $\Phi_{1/2} = 45^\circ$.
- **Fig. 4 (DEB/PEB vs. K):** RMS-DEB and RMS-PEB vs. K for three semi-angles at half power, quantifying the accuracy–latency trade-off.

- **Fig. 5 (DEB/PEB vs. SNR):** Bounds vs. SNR confirming the theoretical $1/\sqrt{\text{SNR}}$ scaling, with $K=5$ bold curve and $K \in \{3, \dots, 9\}$ shaded band.
- **Fig. 6 (DF angular CDF):** Empirical CDF of per-position angular RMSE, solid (vertical Rx) vs. dashed (random tilt), validating $\mathbf{n}_r$ -independence.
- **Fig. 7 (Latency):** Computation time comparison between WLS/GLS at μs -level vs. NLS at ms-level.
- **Fig. 8 (3D positioning CDF):** Empirical CDF of the 3D positioning error for $K \in \{3, 5, 9\}$, comparing all estimators against the PEB and the $K=3$ SVD baseline.
- **Fig. 9 (3D estimates):** Actual GLS position estimates ($K=5$) across three heights ($z=0, 0.6, 1.2$ m), showing the spatial structure of the positioning error.
- **Fig. 10 (Estimator RMSE vs. SNR):** Realized estimator performance vs. SNR for $K=5$, confirming that all estimators track the $1/\sqrt{\text{SNR}}$ slope at constant-factor offsets from the bounds.

Fig. 9 is the only figure that shows the actual output of the GLS estimator (estimated positions vs. references) across the full 3D volume. This validates the spatial behavior of the GLS estimator and visually demonstrates the centimeter-level accuracy reported in Table VI—information that the theoretical DEB/PEB bounds (Figs. 2–5) cannot convey. The former Fig. 9 from the original submission (PEB heat map) has been superseded by the more comprehensive Figs. 2–5 in the revised manuscript, which together provide a far richer characterization of the system's performance.

10. The system relies on sequential measurements over K beam orientations. Although the computational complexity is low, practical beam steering and sampling inevitably introduce latency. If the user moves during the K measurement intervals, the position $\mathbf{r}$ may change between samples, violating the static model assumption in Eq. (12).

This is a pertinent concern, which we have tried to address in two ways:

1. **Computational latency (Section VII-B-3, Fig. 7):** WLS: 0.018 ms median; GLS: 0.025 ms; NLS: 0.65 ms.
2. **Acquisition latency (Conclusion):** For $K = 5$ with state-of-the-art MEMS mirrors (kHz-rate), 6 measurements complete within a few ms. At pedestrian speed (~ 1.4 m/s), displacement during 6 ms is ~ 8.4 mm—below centimeter positioning accuracy. The static model holds for typical indoor mobility.

For higher mobility, future work (ii) proposes intelligent refresh strategies exploiting the previous direction estimate to reduce required orientations.

Section VII-B-3 has thus been added with Fig. 7 (latency), and the Conclusion (Section VIII) states: “For $K = 5$ with state-of-the-art steering mechanisms, the total acquisition window is on the order of milliseconds, resulting in sub-centimeter displacement at pedestrian speeds”.

11. The simulations only compare the proposed NL, GLS, and WLS estimators, without including existing methods or a common baseline such as maximum likelihood estimator. In addition, performance via SNR should also be provided.

(a) Regarding comparison with existing methods. Our system is the first single-LED, single-PD, beam-steered 3D OWP architecture (Table I), so no directly comparable method exists under identical hardware assumptions. We include two baselines in the comparison:

- **Singular value decomposition (SVD) baseline [26]:** This is the only prior single-LED/single-PD beam-steered positioning method ($K = 3$, 2D, unweighted). We extend it to 3D. Table VI shows GLS at $K = 5$ achieves 2.52 cm RMSE vs. 11.05 cm for [26]—a 4.4× improvement.
- **DEB and PEB:** The information-theoretic lower bounds for direction finding and 3D positioning, respectively, serving as the ultimate performance references.

Beyond these baselines, the revised manuscript now includes a **comprehensive standalone direction-finding evaluation** (Section VII-A, entirely new) with dedicated metrics, which was absent from the original submission. This evaluation covers 1,792 positions $\times$ 1,000 Monte Carlo trials (≈ 1.8 million evaluations per estimator) and reports: (i) sub-degree angular RMSE for all estimators, i.e., DEB: 0.52°, NLS: 0.63°, GLS: 0.66°, WLS: 0.77° (Table III); (ii) CDF 90% below 1.14° for all methods (Fig. 6); (iii) robustness to receiver tilt with <3% degradation (Table III); and (iv) μ s-level latency for GLS/WLS (Fig. 7). This standalone evaluation, focused on the direction-finding stage, is arguably the most complete characterization of single-LED direction estimation available in the literature.

Regarding the MLE, it is important to note that **GLS is actually the MLE of the linearized ratio model**, since (42) minimizes the Mahalanobis norm with optimal Σ_β^{-1} . NLS is not the MLE (see the response to Comment 13 of Reviewer 2). A joint MLE over all $K + 1$ measurements is not implementable because measurement $K + 1$ is adaptive (depends on the estimated direction), violating the standard likelihood framework.

(b) Regarding SNR analysis. The revised manuscript provides SNR analysis at two complementary levels:

1. **Theoretical bounds vs. SNR (Fig. 5, Section IV):** RMS-DEB and RMS-PEB vs. SNR for $K \in \{3, \dots, 9\}$, confirming the $1/\sqrt{\text{SNR}}$ CRLB scaling. This shows the best *achievable* performance.
2. **Realized estimator performance vs. SNR (Fig. 10, Section VII-C, entirely new):** Full Monte Carlo simulation of GLS, WLS, and NLS over SNR $\in [10, 50]$ dB (1,792 testbed positions, 1,000 trials per position per SNR point—totaling ≈ 16.1 million estimator evaluations per method). This shows what the proposed estimators *actually achieve*. Key findings:
 - All estimators follow the $1/\sqrt{\text{SNR}}$ slope, confirming asymptotic-regime operation across the entire range.
 - NLS tracks the DEB within a factor $\approx 1.00 \times$ (near-efficient), achieving 0.28° angular RMSE and 1.10 cm positioning RMSE at 20 dB.
 - GLS remains at $\approx 1.3 \times$ and WLS at $\approx 1.6 \times$ above the DEB—constant-factor gaps that persist at all SNR levels, confirming the differences are structural.
 - The SNR axis maps directly to ambient light levels: higher ambient illuminance increases σ^2 , shifting the operating point leftward.

Figure 9 directly responds to this comment by providing estimator performance (not just bounds) across SNR, and closes the gap between the theoretical analysis (Fig. 5) and the single-point results (Table IV).

We have therefore added Section VII-A (direction-finding evaluation), which is entirely new, along with Table II, Table III, and Fig. 6. Fig. 5 (bounds vs. SNR), that have been added in Section IV. Section VII-C and Fig. 10 (estimator RMSE vs. SNR) have also been added in the revised manuscript, whereas Fig. 8 and Table IV include [26] baseline.

12. In Table IV, the meanings of "RMSE," "CDF 90%," and "APE" of CRLB are unclear. RMSE and APE are statistical metrics obtained from multiple Monte Carlo trials, whereas the CRLB represents a theoretical performance bound for unbiased estimators and should not be treated as a random statistic.

The reviewer is correct that the PEB is deterministic at each position. However, it *varies spatially* across 1,792 positions because it depends on $\mathbf{r}$ through the FIM. The metrics for PEB rows are therefore spatial statistics:

- “RMSE”: $\sqrt{\{(1/|\mathcal{R}|)\sum \text{PEB}^2(r)\}}$, is the deterministic spatial RMS, not Monte Carlo.
- “CDF 90%”: 90th percentile of PEB across positions.
- “APE”: spatial mean $(1/|\mathcal{R}|)\sum \text{PEB}(r)$.

A note has been added below Table IV: “For the PEB rows, ‘RMSE’ denotes the root-mean-square of the PEB over all testbed positions (a deterministic spatial statistic, not a Monte Carlo estimate); ‘CDF 90%’ is the 90th percentile across positions; ‘APE’ is the spatial mean PEB”. The term “CRLB” has also been replaced by “PEB” in the table.

13. Section VII-B-3) states that the NL estimator is implemented using Gauss-Newton iterations. Since Gauss-Newton is a typical implementation of MLE under Gaussian noise, it is expected to achieve the CRLB asymptotically. The observed performance gap requires further explanation.

In line with the response to Comment 11 from Reviewer 2, it is important to note that the NLS is not the MLE and is therefore not expected to achieve the CRLB/PEB. The reasons are:

1. **NLS does not fit the full likelihood.** It operates on max-normalized powers (24), which eliminates η (and hence distance information) by design. The MLE would jointly process all $K + 1$ raw powers to estimate $(\mathbf{n}_d, d)$ simultaneously. Our NLS estimates only the direction from K normalized measurements.
2. **Two-stage decomposition loss.** By decomposing direction finding and distance recovery into separate stages, the absolute power information (which carries distance information) is discarded during direction finding. The PEB assumes joint estimation from all $K + 1$ measurements.
3. **PEB assumes perfect beam alignment.** In the $(K + 1)$ -th measurement, the PEB assumes $\cos \phi = \cos \psi = 1$ exactly, which no practical estimator can achieve since it relies on the estimated (imperfect) direction.
4. **Implementation:** The NLS in the revised manuscript is solved via a spherical-coordinate reparameterization (eliminating the unit-sphere constraint) using MATLAB’s `lsqnonlin`

(Levenberg–Marquardt), which exploits the sum-of-squares structure of the cost function through the Gauss–Newton Hessian approximation $\mathbf{J}^T\mathbf{J}$, achieving quadratic convergence near the zero-residual solution.

The performance gap with respect to the PEB (3D positioning bound) is therefore structural (inherent to the two-stage architecture) rather than algorithmic. Crucially, when examining the direction-finding stage in isolation (which is the primary contribution), the NLS estimator is near-efficient: it tracks the DEB within a factor $\approx 1.00\times$ across the entire SNR range (Fig. 10(a), Section VII-C). This confirms that the Levenberg–Marquardt solver with spherical reparameterization converges to the true minimum of the NLS cost function, and that the remaining gap with the PEB originates exclusively from the two-stage decomposition (direction finding + distance recovery), not from the direction-finding algorithm itself.

For 3D positioning (NLS: 2.40 cm vs. PEB: 1.64 cm, factor 1.46 $\times$, Table IV), the additional loss arises from the distance-recovery stage as described above. The text now states: “The performance gap arises from: (i) the two-stage architecture discards distance information during direction finding; and (ii) the PEB assumes perfect beam alignment in the distance-recovery measurement, which no practical estimator can achieve”.

Section V describes the spherical reparameterization and the Levenberg–Marquardt solver (`lsqnonlin`). Section VII-B explains the two sources of the PEB gap. Section VII-C (Fig. 10) confirms NLS tracks the DEB across $\text{SNR} \in [10, 50]$ dB.

14. Several issues in language usage should be corrected: (a) “They presumes...” should be “They presume...”; (b) The sentence beginning with “Therefore, there is a need for dedicated indoor position systems (IPS), which use...” is grammatically broken; (c) “transformer” should be capitalized as “Transformer.”

All three issues have been corrected in the revised manuscript:

- (a) “They presumes” → “They presume” (corrected).
- (b) The sentence has been rewritten for clarity: “Dedicated indoor positioning systems (IPS) are therefore required, and radio-frequency (RF) techniques [...] are widely studied and deployed”.
- (c) “transformer” → “Transformer” when referring to the model architecture (corrected).

All grammar corrections have been applied in Section I of the revised manuscript.

Responses to Reviewer 3

We sincerely thank the reviewer for the constructive and detailed evaluation, and hope that these responses and the corresponding revisions adequately address all of Reviewer 3's concerns.

1. *The authors should summarize and highlight the specific features of the beam-steered single-LED positioning problem, as research in optical wireless localization is already extensive and this distinction is central to the work's significance and motivation.*

We fully agree that the distinguishing features of the single-LED beam-steered paradigm must be made explicit. In the revised manuscript, the Introduction (Section I) has been completely rewritten to clearly articulate these features. Specifically:

- **Infrastructure minimalism:** A single LED replaces the dense multi-LED arrays (3–4 simultaneously visible anchors) required by conventional OWP, eliminating the need for unique identifiers or modulation per lamp and reducing deployment cost (fourth paragraph of Section I).
- **Temporal diversity replacing spatial redundancy:** The K sequential beam orientations create temporal diversity at the transmitter that replaces the spatial diversity traditionally provided by multiple LEDs or multiple PDs. This is a fundamentally different information structure (seventh paragraph of Section I).
- **Receiver-side simplicity:** A single PD with no rotation, no camera, and no PD array is sufficient. The direction-finding stage is mathematically independent of the receiver orientation $\mathbf{n}_r$ (Section VI-C), so no pose calibration is needed.
- **NIR operation:** Decoupling from visible-light illumination constraints allows unrestricted beam steering without affecting room lighting (sixth paragraph of Section I).
- **Ratio-based direction finding:** By forming power ratios, the unknown distance, transmitted power, and all receiver-dependent terms cancel, yielding linear constraints on the direction vector. This is a unique analytical property of the single-LED beam-steered setting.

Table I further contextualizes these features by comparing our system against 12 representative single-LED OWP works, showing that all prior approaches compensate for the single-source limitation at the receiver side (PD arrays, continuous rotation, sensor fusion), whereas ours shifts the burden to the infrastructure via beam steering.

The entire Introduction (Section I) has been rewritten to clearly highlight these distinguishing features. Table I has been updated to include our system with a detailed comparison.

2. *If unique characteristics are not sufficiently distinct from existing multi-LED or multi-PD schemes, the authors should strengthen the motivation to clearly justify the transition to a single-source steering architecture.*

The motivation is now explicitly articulated in the revised Introduction through three complementary arguments:

1. **Deployment cost and scalability:** Multi-LED OWP requires installing and maintaining dense anchor networks with unique identifiers, wiring, and synchronization. A single

steerable LED concentrates all complexity in one infrastructure node, dramatically reducing installation burden (Section I, fourth paragraph).

2. **Availability constraints:** In many indoor environments (warehouses, industrial floors, corridors), luminaire density is insufficient to guarantee 3–4 simultaneous LED visibility—the minimum for multi-LED 3D positioning. A single-LED system eliminates this geometric constraint (Section I, fourth paragraph).
3. **Technological readiness:** Recent advances in MEMS micromirror gimbals have made high-speed beam steering practical for indoor OWC at kHz rates (Section I, sixth paragraph, with references to [36]). A recent comparative study [40] further demonstrates that beam-steered transmitters substantially outperform fixed-antenna configurations in communication coverage, motivating the exploitation of the same capability for positioning.

The distinction from multi-PD schemes is equally clear: systems using 4 tilted PDs [17], [19] or continuously rotating PDs [20], [24], [27] place hardware complexity on the user device, limiting form factor and increasing cost per receiver. Our approach keeps the receiver as simple as possible (single PD, no rotation during direction finding) while shifting diversity to the infrastructure.

Section I now includes the three-argument motivation structure described above, with supporting references and the updated Table I comparison.

3. *The authors should provide a detailed sensitivity analysis regarding the impact of synchronization errors between the transmitter's steering timing and the receiver's sampling intervals.*

We thank the reviewer for this practical concern. In the proposed architecture, the transmitter steers sequentially through K known orientations, and the receiver collects N power samples per orientation. Synchronization requires the receiver to know which orientation is active during each sampling window. We note the following:

1. **Timing structure:** The transmitter follows a predetermined periodic scan pattern (K orientations per cycle). With state-of-the-art MEMS mirrors operating at kHz rates [36], each orientation dwell time is on the order of 1 ms. The receiver simply needs to associate each batch of N samples with the correct orientation index.
2. **Robustness of time-averaged measurements:** The estimators operate on sample means (5). Provided the majority of the N samples within each window correspond to the intended orientation, small timing offsets (i.e., a few samples acquired during transitions) are averaged out. This inherent averaging provides robustness to jitter.
3. **Standard synchronization mechanisms:** Frame-level synchronization between transmitter and receiver is a well-studied problem in OWC systems [11]. Pilot symbols, header sequences, or a simple periodic beacon can achieve the required alignment without adding significant overhead.

A detailed Monte Carlo analysis of synchronization errors under various jitter distributions is an interesting direction. However, given the scope of the present contribution, which focuses on the theoretical framework, bound derivation, estimator design, and direction-finding analysis, and

the availability of established OWC synchronization techniques, we believe that a full synchronization sensitivity study is best addressed in future experimental work, where real hardware timing impairments can be characterized directly.

The Conclusion (Section VIII) now mentions experimental validation as future work item (i), which naturally encompasses synchronization characterization with real hardware.

4. *The authors should supplement the current results with an analysis of positioning performance under different indoor reflection coefficients to evaluate the robustness of the model-based approach against multipath interference.*

We appreciate this comment. The current model considers only the LOS channel, which is the standard approach in model-based OWP literature [9], [29] and [42]. We justify this choice as follows:

1. **Near-infrared (NIR) wavelength:** The system operates at NIR wavelengths, where wall reflectivity is generally lower than in the visible spectrum, reducing non-LOS (NLOS) contributions.
2. **Directional steering:** The beam-steered LED produces a directional beam (Lambertian order m corresponding to $\Phi_{1/2} = 45^\circ$), which concentrates the transmitted power along the LOS path. The ratio of NLOS to LOS power is therefore small for most receiver positions within the room, particularly near the center.
3. **Ratio-based cancellation:** The GLS/WLS estimators rely on *power ratios* between orientations. Since NLOS components (dominated by first-order wall reflections) change slowly with the transmitter orientation compared to the LOS component, the ratio-based formulation provides a degree of inherent robustness to multipath, as the NLOS contribution partially cancels in the ratio.

A systematic study of multipath sensitivity under varying reflection coefficients and room geometries would require integrating a frequency-domain or recursive NLOS channel model, which substantially increases the computational burden and is beyond the scope of the present contribution. We agree, however, that this is an important practical concern and have added it as a future work direction.

The Conclusion (Section VIII) now includes experimental validation as future work item (i), which will naturally address multipath effects in real environments. The LOS assumption is standard in the OWP literature and is explicitly stated in Section II-A.

5. *The authors should include an experimental or simulation-based analysis of the system's performance when the receiver is not perfectly horizontal, as tilt is a common factor in practical mobile scenarios.*

We are pleased to report that this analysis is already included in the revised manuscript as a major new contribution. Section VII-A-2 (“Robustness to Receiver Tilt”) presents a comprehensive Monte Carlo evaluation under random receiver tilts:

- **Tilt model:** θ_{tilt} drawn from a half-normal distribution with $\sigma = 5^\circ$ truncated at 30° ; azimuth φ_{tilt} uniform on $[0^\circ, 360^\circ)$.

- **Scale:** 1,792 positions $\times$ 100 independent random tilts per position $\times$ 1,000 noise trials per tilt = 179.2 million evaluations per estimator.
- **Results (Table III):** All estimators exhibit degradation below 3%, i.e., GLS: +2.24%, WLS: +2.62%, NLS: +1.68%, DEB: +1.69%.
- **Visual confirmation (Fig. 6):** The empirical CDFs under random tilt (dashed curves) are virtually indistinguishable from the vertical-receiver baseline (solid curves).

This near-zero degradation is not coincidental but follows from the analytical proof of receiver-orientation independence in Section VI-C: all three estimators are mathematically independent of $\mathbf{n}_r$ by construction (the common factor η cancels in the ratio-based and max-normalized formulations). The residual <3% degradation is attributable solely to SNR reduction, as a tilted PD intercepts less optical power ($\cos \psi < 1$), lowering the effective SNR, rather than to structural bias.

Section VI-C now provides the analytical proof, whereas Section VII-A-2 presents the Monte Carlo validation with Table III and Fig. 6.

6. *The authors should analyze the impact of different ambient light power levels on the direction estimation stage to quantify the operational limits of the NIR-based positioning link.*

In our model, the noise variance σ^2 lumps together all receiver noise sources: thermal noise and ambient-light-induced shot noise (both modeled as AWGN, as is standard in the OWC literature [42]). Varying the ambient light level is therefore equivalent to varying σ^2 , which in turn changes the SNR. The revised manuscript addresses this concern through two complementary analyses:

1. **Theoretical bounds vs. SNR (Fig. 5, Section IV):** RMS-DEB and RMS-PEB plotted vs. SNR for $K \in \{3, \dots, 9\}$ ($\Phi_{1/2} = 45^\circ$). Both bounds decrease as $1/\sqrt{\text{SNR}}$, confirming the CRLB scaling. At SNR = 14 dB, the $K = 5$ set achieves RMS-DEB $< 0.3^\circ$ and RMS-PEB < 1 cm.
2. **Realized estimator performance vs. SNR (Fig. 10, Section VII-C):** Full Monte Carlo simulation of GLS, WLS, and NLS over SNR $\in [10, 50]$ dB ($K = 5$ with 1,792 positions and 1,000 trials/position). All estimator curves follow the $1/\sqrt{\text{SNR}}$ slope, and the NLS direction estimator tracks the DEB within a factor $\approx 1.00\times$. This demonstrates that at lower SNRs (i.e., higher ambient light), the estimators degrade *gracefully and predictably*, with performance fully characterized by the SNR.

Together, Figs. 5 and 10 provide a complete parametric characterization: Fig. 5 shows the best achievable performance (bounds), while Fig. 10 confirms that the proposed estimators track those bounds at constant offsets across all ambient light conditions. For a specific environment, the reader can compute σ^2 from standard shot-noise formulas [42] and read the expected performance directly from these figures.

Figure 5 (bounds vs. SNR) has been added in Section IV. Section VII-C and Fig. 10 (estimator RMSE vs. SNR) have also been added, showing realized performance across SNR $\in [10, 50]$ dB.

7. *The authors should provide a more comprehensive comparison of the genetic algorithm's convergence efficiency against other global optimization heuristics to justify its selection for orientation-set design.*

We chose the GA for the following reasons, which are now discussed in Section IV:

1. **Non-convex, multimodal landscape:** The DEB objective (22) involves powers of cosines, distance-dependent factors, and the FOV constraint, producing a non-smooth, multimodal landscape. Gradient-based methods (e.g., SQP, interior-point) are prone to local minima; population-based heuristics are better suited.
2. **Bounded, continuous search space:** The decision variables (tilt–azimuth pairs) lie in compact domains ($\theta_i \in [0^\circ, 80^\circ]$, $\phi_i \in [0^\circ, 360^\circ]$). GAs with real-valued encoding handle such bounded continuous problems naturally.
3. **MATLAB integration:** MATLAB’s Global Optimization Toolbox provides a mature, well-tested GA implementation with built-in parallel evaluation, adaptive feasible mutation, and early-termination criteria, facilitating reproducibility.
4. **Empirical quality of solutions:** As shown in Table III, the GA-optimized sets exhibit physically interpretable and symmetric patterns (e.g., $K = 4$: four orientations at 29.9° tilt, uniformly spaced by 90° in azimuth—the intuitively optimal configuration). Figure 2 further demonstrates that the GA consistently outperforms 50 random realizations per K , reducing the median DEB by 49%–72%.

We acknowledge that a systematic comparison with alternative metaheuristics (particle swarm optimization, differential evolution, simulated annealing) would be informative. However, the orientation-set design is an *offline* one-time computation (not repeated per measurement), so absolute convergence speed is not a critical bottleneck. The quality of the resulting sets—demonstrated by their monotonically decreasing DEB ($1.30^\circ \rightarrow 0.36^\circ$), physically interpretable patterns, and consistent superiority over random sets—provides strong empirical evidence that the GA produces high-quality solutions. A formal comparison of metaheuristics for this specific problem is an interesting algorithmic study that we plan to pursue in future work.

Section IV now includes a paragraph justifying the GA choice. Table III and Fig. 2 provide empirical validation of solution quality.

8. *The authors should clarify the physical constraints of the mechanical steering system and analyze how the maximum angular velocity of the beam-steerer affects the overall localization latency.*

We address this through the new latency analysis in Section VII-B-3 and the Conclusion:

- **Computational latency (Section VII-B-3, Fig. 7):** For $K = 5$ with powers pre-acquired, GLS completes in 0.025 ms (median) and WLS in 0.018 ms. NLS requires 0.65 ms due to the iterative Levenberg–Marquardt solve. The closed-form estimators are therefore negligible contributors to overall latency.
- **Steering latency:** The total pipeline requires $K + 1$ sequential measurements. The dominant latency contributor is the physical steering time between orientations. With state-of-the-art MEMS micromirror gimbals achieving angular velocities on the order of 10^4 – 10^5 °/s [36], a typical 60° reorientation takes ~ 0.6 – 6 ms. For $K = 5$, the six beam transitions and sampling windows result in a total acquisition time on the order of a few milliseconds to tens of milliseconds, depending on the specific hardware.

- **Impact on user motion:** At pedestrian speed (~ 1.4 m/s), a 6 ms acquisition window corresponds to ~ 8.4 mm displacement, well below the centimeter-level positioning accuracy (Table IV). Even with slower steering mechanisms (tens of ms total), the displacement remains within the system's accuracy budget.

We note that the proposed framework is agnostic to the specific steering technology, provided the beam-steerer preserves the native LED emission pattern. The key requirement is that it can reach the K predefined orientations with sufficient accuracy and speed relative to the user's mobility. A detailed hardware characterization is planned as part of the experimental validation (future work item (i)).

Section VII-B-3 provides the computational latency analysis with Fig. 7. The Conclusion states: “For $K = 5$ with state-of-the-art steering mechanisms, the total acquisition window is on the order of milliseconds, resulting in sub-centimeter displacement at pedestrian speeds”.

9. *The authors should analyze the Fisher Information Matrix rank conditions for cases where $K \geq 3$ but the chosen orientations are nearly coplanar, as this affects the practical deployment rules.*

This is an insightful observation. Section III-E (“Number of LED Orientations”) already discusses the three FIM regimes:

- **Underdetermined ($K \leq 2$):** The FIM is rank-deficient and the CRLB diverges.
- **Just-determined ($K = 3$):** “With three distinct, non-coplanar LED orientations plus the range measurement, the total of four gradient vectors can span the 3D parameter space” (emphasis added). The text explicitly notes that the absence of redundancy makes the bound “highly sensitive to small perturbations in orientation geometry”.
- **Overdetermined ($K \geq 4$):** The FIM accumulates redundant outer products, improving conditioning and reducing the PEB.

The near-coplanar case the reviewer raises corresponds to the degenerate $K = 3$ scenario where the gradient vectors become nearly linearly dependent, causing the FIM to become ill-conditioned (large condition number). In the limit of exact coplanarity, the FIM loses rank and the CRLB diverges in the direction orthogonal to the common plane. This is precisely why Section III-E stresses the non-coplanarity requirement and why the GA optimization is essential: **the GA inherently avoids degenerate configurations** because they yield very large DEB values and are therefore penalized by the objective function (22).

This is also visible in the optimized orientation sets (Table III): for $K \geq 5$, the GA consistently places orientations at two distinct tilt levels (one near 0° and one near 65° – 70°), ensuring that the gradient vectors span three dimensions. The DEB heat maps (Fig. 3) further demonstrate that the optimized configurations maintain bounded DEB across the entire testbed, confirming that the GA avoids ill-conditioned geometries.

Section III-E already discusses the three regimes, including the non-coplanarity requirement for $K=3$. The GA optimization naturally enforces well-conditioned FIM through the DEB objective.

10. *The authors should analyze the error propagation from the direction-finding stage to the distance-recovery stage to explain the total system error budget more clearly.*

We agree that understanding the error budget is important. The revised manuscript addresses this through two complementary mechanisms:

1. **Theoretical analysis (Section III-D, Eq. 21):** The PEB is derived from the full $(K + 1)$ -observation FIM, which accounts for all K direction-finding measurements plus the distance-recovery measurement. Comparing the DEB (direction-finding only) with the PEB (full pipeline) reveals the relative contribution of each stage. For $K = 5$: RMS-DEB = 0.52° (Table III), RMS-PEB = 1.64 cm (Table IV). The DEB translates to ~ 1.1 cm direction-induced displacement at $d \approx 1.2$ m, which dominates the PEB budget.
2. **Monte Carlo evidence (Section VII-B, Table VI):** The performance gap between the estimators and the PEB is explicitly attributed to two sources: “(i) the two-stage architecture discards distance information contained in the absolute power levels during direction finding, whereas the PEB assumes joint estimation from all $K + 1$ measurements; and (ii) the PEB assumes perfect beam alignment in the distance-recovery measurement, which no practical estimator can achieve”.

In other words, the direction-finding error propagates to the distance-recovery stage in two ways: (a) the direction estimate $\hat{\mathbf{n}}_d$ determines the beam-alignment quality for the $(K+1)$ -th measurement (imperfect alignment reduces $\cos \phi$ and $\cos \psi$ below unity, lowering the ranging SNR); and (b) the estimated distance $\hat{d}$ is multiplied by $\hat{\mathbf{n}}_d$ to produce the final position, so angular errors are amplified by the distance. These effects are visible in Fig. 9, where errors grow at larger distances from the transmitter (boundary and corner positions).

Section III-D defines the PEB including the distance-recovery measurement. Section VII-B (text below Table IV) explicitly discusses the two sources of error propagation.

11. The authors should ensure that all notation and variable definitions in Section II-A are strictly consistent with the mathematical expressions used in the Appendices.

We thank the reviewer for this attention to detail. We have carefully reviewed all notation in Section II-A and verified its consistency with the Appendices. Specifically:

- **Appendix A, i.e., proof of (35):** Uses $\hat{\mu}_i, \mu_i, n_i, \beta_i$ and the function $g(x_1, x_2) = \left(\frac{x_2}{x_1}\right)^{1/m}$, which are all defined consistently with Section II (4)–(5) and Section VI-A (30)–(35).
- **Appendix B, i.e., proof of (38):** Uses $\tilde{n}_i = \hat{\beta}_i - \beta_i$ as defined in (36), with variance and covariance expressions matching (37)–(38) exactly.
- **Notation paragraph:** A comprehensive notation summary (Section I, “Notation” paragraph) defines all conventions (bold lowercase for vectors, bold uppercase for matrices, hat for estimates, overbar for sample means, etc.) used throughout the paper and appendices.

We have verified that all symbols are introduced before use and that no notation conflicts exist between the main text and the appendices.

Notations have been verified for consistency. The Notation paragraph in Section I provides a comprehensive reference for all conventions used throughout the paper.